

21EIC552J – Industrial Process Automation

Question Bank

MODULE – I

1. List the evolution of automation technology over the past decades
2. Define Physical process and name its categories
3. What is a process signal and list the types of physical signals.
4. List the benefits of Automation
5. Draw the basic structure of automation system
6. Draw the Typical automation cycle.
7. What are the tasks performed by Control Subsystem
8. Describe Process Management system through the schematic and operation of the unattended/ attended/ automated water heater system.
9. Explain briefly about the automation steps in water-heating process
10. Discuss in detail about Input, output and human interface Instrumentation Subsystem with an example
11. Draw the structure of analog instrumentation device and explain its measurement and control mechanism.

MODULE – III

Explain briefly about various functional elements of SCADA system
Describe how MTU handles analog control instructions to open and close a valve in a SCADA system.
Describe briefly about the architecture of a DCS
Explain the design considerations and requirements of operator interface in detail.

Explain briefly about the analog input and output module of a SCADA system
Describe how MTU handles discrete control instructions to open and close a valve in a SCADA system.
Explain the architectural parameters of LCU and compare its different configurations.
Describe briefly about the hybrid system architecture

Explain briefly about the digital input and output module of a SCADA system

Discuss briefly about the functional elements of a Remote Terminal Unit (RTU)
Explain the function block programming approach with an example
Describe briefly about the architecture of a DCS